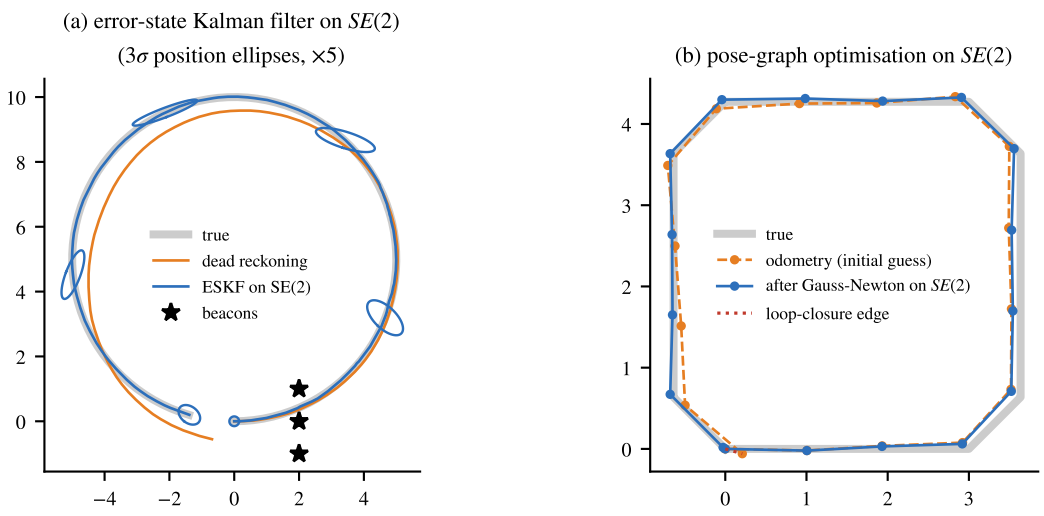
(a) error-state Kalman filter on $SE(2)$
(3σ position ellipses, ×5)
true
dead reckoning
ESKF on SE(2)
beacons
10
8
6
4
2
0
−4
−2
0
2
4
(b) pose-graph optimisation on $SE(2)$
true
odometry (initial guess)
after Gauss-Newton on $SE(2)$
loop-closure edge
4
3
2
1
0
0
1
2
3